

EE P 595 Project Update — Progress Report

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Current Status

The pipeline is up. I have Texas-100X loading end-to-end (924,954 rows, 100 procedure classes after the top-K filter, 441 hospitals, THCIC_ID excluded from model inputs as the proposal promised), teacher and student training validated on a real run, and all three MIAs from the proposal (loss-based (Yeom et al.), shadow models (Shokri et al.), and an approximate LiRA) implemented and producing results. I also have a full LiRA implementation with 64 reference models, because the approximate version was clearly leaving a signal on the table once I re-read Carlini et al. carefully. Each run writes into a timestamped experiment folder so I can compare configurations without overwriting prior results.

Here are some numbers from a 50k/10k/50k train/val/test split:

Model	Attack	AUC	TPR @ 1% FPR	Train / Test Acc
Teacher	Loss-based	0.642	1.2%	63.6% / 47.4%
Teacher	LiRA (approx)	0.652	1.8%	63.6% / 47.4%
Student	Loss-based	0.510	1.0%	32.8% / 33.9%
Student	LiRA (approx)	0.509	0.9%	32.8% / 33.9%

This looks like the textbook "distillation is good for privacy" story, where the teacher overfits with a 16-point gap, the attacker catches that gap, the student is small and underfits, and the attacker is reduced to flipping a coin. The subgroup analysis was quite interesting and where I think I would like to push this project towards. Stratifying the same attack scores by per-sample teacher loss, the worst loss quartile still has AUC 0.88 and ~75% TPR @ 1% FPR on the student, versus 0.95 / 87% on the teacher. The student's average AUC of 0.51 is being held down by easy, common, well-classified samples that were never very vulnerable in the first place. The samples

that were memorized (the rare conditions, the long-tail patients, the cases the teacher overfit on) keep almost all of their leakage after distillation. The class-frequency stratification tells the same story, where rare-class samples (<1% of training data) sit at AUC 0.68 on the student versus 0.27 for common classes.

That asymmetry is the point I want to lead this project. Essentially, showing that knowledge distillation does not make a model more private, rather it makes the average of the model more private while leaving the most vulnerable patients almost as exposed as before. The framing I want to build the rest of the project around is the same one Bagdasaryan et al. (2019) used for differential privacy, where a privacy-motivated technique whose benefit accrues to the population that was already safe is not actually doing the job it advertises. As far as I can tell, this disparate-impact framing has been applied carefully to DP and to federated learning, but not to knowledge distillation, and the existing KD-privacy literature reports almost exclusively population-level AUC.

Challenges/Struggles

I think the main thing that I struggled with is that the current student model underfits. At 34% test accuracy the student isn't really competing with the teacher, which makes the privacy comparison suspect since a healthier student would probably leak more. This is the single biggest threat to the disparate-impact claim, because a critic can reasonably say "you only see the disparity because your student didn't learn enough." I'd like to spend some time fixing this before any external write-up.

Another struggle that I am seeing is that the loss-quartile stratification might be circular. I'm partly stratifying samples by the same loss the attack is built on. The class-frequency and confidence-quantile breakdowns are independent of that and tell the same story, but I need to add at least one external signal (Feldman & Zhang (2020) memorization scores are the cleanest option) before I'd feel comfortable making the disparate-impact claim.

Per-sample provenance plumbing. For LiRA to work and for the subgroup analysis to be honest, every attack score has to be tied back to a specific source row. I had to thread `source_indices` through the dataset, the loaders, and the score writers. Most of my debugging time traced back to this.

Next Steps

The remaining course-project work, in priority order:

- Re-tune the student so its test accuracy lands in the 40–45% range. Without this the disparate-impact comparison isn't credible.
- Wire confidence filtering into the training entry point and run the attack suite on it.
- Generate the privacy–utility plot the original proposal promised, but include a second version of the same plot stratified by loss quartile, so the reader can see the population number and the worst-case number side by side.

The remainder of the quarter will be spent on the above work.